

The update strictly follows scoring, so no point is ever used to predict itself; a covariate-arrival policy declares, per exogenous channel, whether its value at $t + h$ is known at prediction time (a planned promotion) or only revealed with the target (a realized driver). After a warmup of w steps the harness records, as functions of stream position, four trajectories: the rolling scaled error, the rolling empirical coverage, the interval width, and the cumulative compute cost.

B. Metrics

Point accuracy uses the mean absolute scaled error (MASE) [5], which divides the absolute error by the in-sample seasonal-naive error so that values below one beat the naive baseline and the scale is comparable across series. Interval quality is read two ways that cannot be gamed independently: the empirical coverage (the fraction of revealed truths inside the interval, compared to the nominal $1 - \alpha$) and the interval width (a calibrator that only widens to reach coverage is not impressive), and both are summarized by the M4 interval score MSIS [6], which charges width plus a miss penalty per unit so a narrow interval that misses cannot look good. Probabilistic sharpness is scored with the weighted quantile loss, a proper scoring rule [7]. The compute-cost trajectory is the constant-cost-per-step signature that separates a stateful model from a replay-everything wrapper.

C. The *preqts* package

The harness is extracted from the atlas and published as the open-source package *preqts*. It provides the *StatefulForecaster* protocol, the streaming loop with the covariate-arrival policy, the four-trajectory recorder, and the online conformal calibrators of Sec. III. Because the streaming artifact carries only aggregate metric trajectories (no raw series values are recoverable from rolling MASE, coverage, width, or cost), the demonstration ships even for local-only-licensed real datasets whose raw excerpts are otherwise redacted.

III. ONLINE CONFORMAL CALIBRATION

Given a nonconformity score (here the absolute residual), split-conformal prediction issues an interval whose width is a quantile of past scores; under exchangeability its coverage is $1 - \alpha$. Streams are not exchangeable, so the effective level must adapt. **Adaptive Conformal Inference** [2] updates the level each step by the observed miss,

$$\alpha_{t+1} = \alpha_t + \gamma (\alpha - \text{err}_t), \quad \text{err}_t = \mathbf{1}\{y_{t+h} \notin \hat{I}_t\}, \quad (2)$$

so that long-run coverage converges to the target under arbitrary distribution shift. **Conformal-PID control** [3] generalizes this to a proportional-integral-derivative controller on the coverage error; ACI is the integral-only special case, and the proportional and derivative terms buy faster coverage recovery after a regime change. Both wrap any point forecaster and change only the interval, never the point prediction.

TABLE I
STREAMING INTERVAL CALIBRATION AT NOMINAL 80% OVER 15 CASES
(MEAN AND WORST-CASE ABSOLUTE COVERAGE ERROR
| COVERAGE - 0.80 |). ONLINE CONFORMAL CALIBRATION CUTS THE
ERROR 7-10× AND CAPS THE WORST CASE.

Streaming method	mean cov - 0.80	worst case
Seasonal-naive (floor)	0.0599	0.307
Theta, raw intervals	0.0416	0.112
Theta + ACI	0.0057	0.022
Theta + Conformal-PID	0.0060	0.022

IV. THE ATLAS TESTBED

We exercise the harness on ChronoScope, an atlas of fifteen committed cases baked deterministically (each artifact a pure function of case and seed). The cases span distinct regimes by construction, level shift, deterministic chaos (Mackey-Glass), GARCH heteroskedasticity, intermittent demand, long memory (fractional), coupled daily/weekly seasonality, and exogenous promotions, and include real CC-BY series (UCI Electricity, Beijing PM2.5, M4 daily and hourly) and two honesty controls, white noise and a near-random walk, on which a suspiciously large “skill” would flag harness leakage. Each case is scored by the full method ladder (classical, statistical, gradient-boosted, deep, and stateful foundation models such as Chronos [9] and TimesFM [10]) and by the streaming bench of Sec. II at nominal coverage 80%, horizon one, warmup twenty-four. The streaming roster reported here is the honest floor (seasonal-naive), a strong classical live method (Theta [8]) with raw intervals, and that same Theta wrapped in ACI and in Conformal-PID.

V. RESULTS

Table I is the headline. Averaged over the fifteen cases, the raw Theta intervals miss the nominal 80% target by a mean absolute coverage error of 0.042 and the seasonal-naive floor by 0.060, with worst-case per-series errors of 0.11 and 0.31. Wrapping the *same* Theta point forecaster in ACI or Conformal-PID cuts the mean coverage error to 0.0057 and 0.0060, a seven-fold reduction over raw Theta and a ten-fold reduction over seasonal-naive, and caps the worst-case per-series error at 0.022. The point forecast is unchanged (the MASE of Theta, Theta+ACI and Theta+PID is identical per case); only the interval self-corrects.

The effect is regime-robust. Where raw intervals drift most, the calibrators pull coverage back to target: on intermittent demand the raw Theta over-covers at 0.91 while ACI sits at 0.807; on a chaotic Mackey-Glass stream raw Theta under-covers at 0.77 while ACI reaches 0.801; on real UCI electricity raw Theta under-covers at 0.76 while ACI reaches 0.796. On the white-noise and random-walk honesty controls the calibrated coverage holds at 0.79 (against raw 0.75), i.e. the harness does not manufacture skill on unpredictable series, exactly the leakage guard the controls exist to enforce. Because the point forecast never changes, this is a pure calibration gain: the same forecaster, with intervals that stay honest as the stream drifts.

VI. DISCUSSION AND HONEST SCOPE

The finding, that online conformal calibration keeps streaming intervals near nominal under shift, is not by itself surprising: it is what ACI and Conformal-PID were designed to do. The contribution is that it is measured under the correct protocol. A stateless rolling-window harness cannot see a calibrator recover coverage after a regime change, because it does not carry the calibrator’s state across windows; the prequential harness does, and it charges the compute cost that distinguishes a constant-cost stateful method from a replay wrapper. The honesty controls and the identical point forecasts across the raw and calibrated variants rule out the two easy ways such a result could be an artifact (harness leakage; a point-accuracy confound).

Scope. The demonstration uses a Theta base forecaster because it is a strong classical method that runs live; the calibration is method-agnostic and wraps any forecaster, which is the point of the protocol. The atlas mixes seeded synthetic regimes with a few real CC-BY series; the synthetic cases are constructed to isolate regimes and the controls guard against leakage, but the real-data coverage is reported on a handful of public series, not a large corpus. Results are shown at horizon one and nominal 80%; the harness supports other horizons and levels. The value we claim is the reusable stateful-streaming harness and the leakage-guarded measurement, not a new calibrator.

VII. CONCLUSION

We built and released `preqts`, a prequential streaming evaluation harness for stateful time-series forecasters with an explicit covariate-arrival policy, filling a verified gap left by stateless-window benchmarks, and used it to show, artifact-backed and control-guarded, that online conformal calibration reduces streaming interval coverage error roughly seven-to-tenfold and caps the worst case while leaving the point forecast unchanged. Evaluating forecasters the way they are deployed, as a stream with state, is what makes the calibration story visible; the harness and the committed atlas make it reproducible.

DATA AND CODE AVAILABILITY

The harness is the open-source package `preqts` (<https://pypi.org/project/preqts/>). The atlas code, the committed per-case artifacts (trace, analysis, and streaming trajectories) and the interactive workbench are at https://github.com/fsantibanezleal/CAOS_RES_ChronoScope under the MIT license; live at <https://chronoscope.fasl-work.com>. Real datasets (UCI Electricity, Beijing PM2.5, M4 excerpts) ship under CC-BY-4.0 with attribution; streaming artifacts carry aggregate trajectories only.

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